NAZARBAYEV UNIVERSITY
SCHOOL OF ENGINEERING AND DIGITAL SCIENCES

Project Assignment #3

CSCI502/702 – HARDWARE/SOFTWARE CO-DESIGN

Spring Semester 2026

IMU GUI DEVELOPMENT USING QT 6

DUE TIME AND DATE

Monday/Wednesday 16/18 March – class presentation
Wednesday 18 March, 23.59 – report/code in Moodle and GitHub

LEVEL OF COLLABORATION ALLOWED

You will be working in groups of 3-4 students

REFERENCE

1. **Exploring Raspberry PI by Derek Molloy (PDF in Moodle)**
   Note: some Linux commands and packages mentioned in the textbook(s) can be already deprecated. So, searching for alternatives on the web is required sometimes.
2. **Nibedit Dey. *Cross-Platform Development with Qt 6 and Modern C++*, Packt, 2021**
3. **Lee Zhi Eng, Ray Rischpater. *Application Development with Qt Creator. Build cross-platform applications and GUIs using Qt 5 and C++.* 3rd ed. 2020**
4. **Use of generative AI tools is allowed.**

DELIVERABLES REQUIRED

1. **An individual detailed report in PDF format with screenshots of completed tasks and program codes, submitted to the GitHub and Moodle submission folders**
2. **1-3 min video demo of your working application, submitted to the Moodle submission folder together with the PDF report as a zip archive file.**

INTRODUCTION

The term **Internet of Things (IoT)** is widely used to describe the network of devices such as vehicles, and home appliances that contain electronics, software, actuators, and connectivity which allows these things to connect, interact and exchange data. The IoT involves extending Internet connectivity beyond standard devices, such as desktops, laptops, smartphones and tablets, to any range of traditionally dumb or non-internet-enabled physical devices and everyday objects. Embedded with technology, these devices can communicate and interact over the Internet, and they can be remotely monitored and controlled. In this project, you will use the RPi board and an IMU sensor to create an embedded IoT device for communicating sensor orientation information into a graphical user interface (GUI) over network.

Qt (pronounced "cute" or as an initialism) is cross-platform software for creating GUIs as well as cross-platform applications that run on various software and hardware platforms such as Linux, Windows, macOS, Android or embedded systems with little or no change in the underlying codebase while still being a native application with native capabilities and speed. The latest version of the Qt Framework is Qt 6.x.

ASSIGNMENT DETAILS

TASK 1. CREATING A SENSOR WEB SERVER

Following **Chapter 12 of the Exploring Raspberry Pi textbook** configure the **Nginx (or Apache)** web server on the RPi board and create a simple IMU sensor web page for outputting the sensor orientation quaternion readings similarly **to Fig. 12-5** using CGI and/or PhP scripts. Provide screenshots of a web-page accessed from your laptop (client side) in the report for various sensor measurements.

TASK 2: GETTING STARTED WITH QT 6

1. Apply for the free **Qt Educational License for Developers** at https://www.qt.io/qt-educational-license **using your NU email.**

2. Once you follow the Qt account setting steps, you will receive the Qt download instructions. As a reference you may follow installation instructions from **Chapter 1 and Chapter 5 of N. Dey. *Cross-Platform Development with Qt 6 and Modern C++*, Packt, 2021 book** and/or from the YouTube Qt with C++ course. Episode 1

   https://youtube.com/playlist?list=PLUbFnGajtZlXbrbdlraCe3LMC_YH5abao&si=7nvOP3KlFVT1QeDU

3. Learn basics of creating GUI applications with Widgets through following, for example, Qt 6 Crash Course for Beginners https://www.youtube.com/watch?v=cXojtB8vS2E and/or reference textbooks:

   - Nibedit Dey. *Cross-Platform Development with Qt 6 and Modern C++*, Packt, 2021
   - Lee Zhi Eng, Ray Rischpater. *Application Development with Qt Creator. Build cross-platform applications and GUIs using Qt 5 and C++.* 3rd ed. 2020

   YouTube Qt with C++ course.

   https://youtube.com/playlist?list=PLUbFnGajtZlXbrbdlraCe3LMC_YH5abao&si=7nvOP3KlFVT1QeDU is good for learning advanced Qt elements.

4. Think of some interesting (not very complicated) GUI project and implement it using Qt. Examples: calculator **(example is available in Lee Zhi Eng, Ray Rischpater. *Application Development with Qt Creator. Build cross-platform applications and GUIs using Qt 5 and C++*)**, mouse pointer coordinates tracking, etc. The main point of this task is to show you can understand and develop a relatively modest Qt GUI application.
   Demonstrate successful execution of your Qt application on your PC during project presentation session.

TASK 3. IMU GUI DEVELOPMENT

In this task you will develop a client/server application to visualize an IMU sensor orientation on your laptop/PC. The RPi with an interfaced IMU would act as a server and send processed orientation data to your laptop/PC (client). The client PC should run a Qt GUI for visualizing the IMU sensor 3D orientation. Please implement the following:

1. Study **Chapter 12** and the **C++ Client/Server Section** on **pages 523-526 of the Exploring Raspberry Pi textbook**.
2. Study the Qt GUI application development in Chapter 14 of the **Exploring Raspberry Pi textbook**. In particular, focus on the **Remote UI Application Development** section **(Fat-Client Qt GUI Application subsection) on pages 602 – 606 of the textbook.** Example program codes from Chapters 11 and 13 can be downloaded from the textbook GitHub page.
3. Based on your Project 1 and 2 developments implement on the **server side (RPi)** the code with the following functions (reusing Project 1 and 2 code):
   - Reading the IMU raw sensor measurement data and converting it to physical values.
   - Processing IMU sensor data with a sensor measurement fusion algorithm and sending the output sensor orientation quaternion data to the server socket communication for reading by the client side. **Please make the quaternion component normalization before sending the data to the algorithm (Project 2).**
4. On the **client side (your laptop/PC)** you will implement GUI application in the Qt Creator which handles:
   - the client socket communication. It reads the data.
   - runs the data processing and visualizes IMU orientation.

Please develop the IMU sensor orientation visualization in the Qt GUI. You may use a provided in Moodle **test-opengl** framework based on OpenGL and QTthread classes (the framework was designed for Qt 5-based projects, so there might be some errors that need to be fixed). You may use alternative visualization tools, e.g. QtCanvas3D or others, if desired.

You may follow the online C++ GUI with Qt Tutorial videos #28 to #35 (if deemed necessary) from https://www.youtube.com/watch?v=JaGqGhRW5Ks for learning how to work with QThread class in the Qt.

TASK 4. THREADED IMU INTERFACING

Modify your **Project 2** IMU code to implement a **multithreaded server side (RPi)** using C++ threads as follows:

- Thread 1 reads the IMU raw sensor measurement data, converts it to physical values as explained in Steps 1 and 2, and passes it to Thread 2;
- Thread 2 runs the IMU sensor measurement fusion algorithm that accepts the IMU data from Thread 1, and passes the quaternion estimates to Thread 3, the server socket communication, for reading by the client side. Please make the quaternion component normalization before sending the data.
- Thread 3 implements the server socket communication and sends the data to the socket for reading by the client side.

Use shared variables (protected by mutexes if needed) to pass the data from one thread to another.

GRADING CRITERIA

Demonstration and grading of your working RPi&IMU setups and program solutions (with questions/answers) will be done during the class lab session on Monday/Wednesday 16/18 March. Please prepare a detailed report with program code and submit it to the project folder in Moodle and GitHub Classroom on the link https://classroom.github.com/a/T3Q_PJUO by the end of Wednesday 18 March.

This project evaluation will be done using individual grading depending on the level of participation and understanding of the project assignments.

Late lab demonstration/report submission penalty – 5% for the first day after the deadline, later 10% per each subsequent day

	Class Demonstration	**Report**
Task 1.	10 – weak presentation, little understanding, not correctly working system or 15 – clear, competent, correctly working system;	0 – minimal documentation or 5 – clear rigorous, well-structured reporting
Task 2.	10 – weak presentation, no/little understanding of the implemented Qt GUI design, not correctly working system, or 15 – acceptable presentation, correctly working program, average understanding of Qt GUI design, or 30 – clear presentation, good understanding of the implemented program	0 – minimal documentation or 5 – acceptable but some missing sections/descriptions or 10 – clear, rigorous, well-structured reporting
Task 3.	10 – weak presentation, no/little understanding of the implemented C++ threads, not correctly working system, or 15 – acceptable presentation, correctly working program, average understanding of C++ threads, or 25 – clear presentation, good understanding of the implemented program	0 – minimal documentation or 5 – clear, rigorous, well-structured reporting
Teamwork & individual contribution	0 – no/little individual contribution or 5 – good individual contribution. 5 – demonstrated strong teamwork.	0 – No outline of individual student contributions in the report or 5 – Clear outline of individual student contributions in the report
TOTAL	**Max – 75**	**Max – 25**